

BASH Programming

Removing Characters from String in Bash

1 year ago • by Karim Buzdar

At times, you may need to remove characters from a string. Whatever the reason is, Linux provides you with various built-in, handy tools that allow you to remove characters from a string in Bash. This article shows you how to use those tools to remove characters from a string.

The article covers how to perform the following:

- Remove character from string using sed
- Remove character from string using awk
- Remove character from string using cut
- Remove character from string using tr

The commands shown in this article were performed in Ubuntu 20.04 Focal Fossa. The same commands can also be performed on other Linux distributions that have the above tools available. We will use the default Terminal application to run the commands. You can access the Terminal application using the Ctrl+Alt+T keyboard shortcut.

Remove Characters from String Using sed

Sed is a powerful and handy utility used for editing streams of text. It is a non-interactive text editor that allows you to perform basic text manipulations on input streams. You can also use sed to remove unwanted characters from strings.

For demonstration purposes, we will use a sample string and then pipe it to the sed command.

Remove Specific Character from String

Using sed, you can remove a specific character from a string. For example, to remove “h” from the string “*hello, how are you?*” the command would be:

```
$ echo "hello, how are you?" | sed 's/h//'
```

This will only remove the first occurrence of ‘h’ in the string.

```
kbuzdar@ubuntu:~$ echo "hello, how are you?" | sed 's/h//'  
ello, how are you?  
kbuzdar@ubuntu:~$
```

To remove all occurrences of ‘h’ from the string, use the following command:

```
$ echo "hello, how are you?" | sed 's/h//g'
```

Where **g** stands for global. It will remove all occurrences of ‘h’ in the string.

```
kbuzdar@ubuntu:~$ echo "hello, how are you?" | sed 's/h//g'  
ello, ow are you?  
kbuzdar@ubuntu:~$
```

Remove First Character from String

To remove the first character from the string “*hello, how are you?*” the command would be:

```
$ echo "hello, how are you?" | sed 's/^./' file
```

Where (.) matches exactly a single character and (^) matches any character at the beginning of the string.

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Remove Last Character from String

To remove the last character from the string “*hello, how are you?*” the command would be:

```
$ echo "hello, how are you?" | sed 's/.$//'
```

Where (.) matches exactly a single character and (\$) matches any character at the end of the string.

Remove First and Last Character from String

To remove the first and last character from the string “*hello, how are you?*” the command would be:

```
$ echo "hello, how are you?" | sed 's/^./;;s/.$//'
```

Remove Characters from String Using awk

Awk is a powerful scripting language used for pattern matching, along with text processing. Awk allows you to filter and transform text in various ways. You can also use awk to remove characters from strings.

For demonstration purposes, we will use a sample string and then pipe it to the awk command.

Remove First Character From a String

To remove the first character from the string “*hello, how are you?*” the command would be:

```
$ echo "hello, how are you?" | awk '{ print substr( $0, 2 ) }'
```

Where (\$0) is the whole target string and (2) is the character starting position. The above command removes the first character, ‘h,’ character number ‘1,’ and returns the target string beginning with the second character, ‘e.’

Remove First Two Characters from String

You can also remove a specific number of characters from the beginning of a string. For example, to remove the first two characters from the string “*hello, how are you?*” the command would be:

```
$ echo "hello, how are you?" | awk '{ print substr( $0, 3 ) }'
```

The above command will remove the first two characters, ‘he,’ or character numbers ‘1 and 2,’ and returns the target string beginning with character number ‘3,’ or ‘l.’

Remove Last Character from String

To remove the last character from “*hello, how are you?*” the command would be:

```
$ echo "hello, how are you?" | awk '{ print substr( $0, 1, length($0)-1 ) }'
```

Where **length(\$0)-1** means deducting ‘1’ from the total character length.

The above command will print the string beginning with character number ‘1’ up to **length(\$0)-1** to strip off the last character.

There are ‘19’ characters (including spaces) in the above string. The command will work by printing all characters, starting with character ‘1’ and up to character ‘18,’ while removing the last character ‘19.’

Remove Last Two Characters from String

To remove the last two characters from “*hello, how are you?*” the command would be:

```
$ echo "hello, how are you?" | awk '{ print substr( $0, 1, length($0)-2 ) }'
```

Where **length(\$0)-2** means deducting ‘2’ from the total character length.

The above command will print the string, beginning with character number ‘1’ and up to character number ‘**length(\$0)-2**,’ to remove the last two characters in the string.

Remove Both First and Last Characters from String

To remove both the first and the last characters from the string “*hello, how are you?*” the command would be:

```
$ echo "hello, how are you?" | awk '{print substr($0, 2, length($0) - 2)}'
```

Where **length(\$0)-2** means deducting ‘2’ from the total character length.

The above command will print the string, beginning with character number ‘2’ up to character number ‘**length(\$0)-2**,’ to remove the first and last character.

Remove Character from String Using cut

Cut is a command-line tool commonly used to extract a portion of text from a string or file and print the result to a standard output. You can also use this command for removing characters from a string.

For demonstration purposes, we will use a sample string and then pipe it to the cut command.

Remove First Character from String

To remove out the first character from the string, “*hello, how are you?*” the command would be:

```
$ echo "hello, how are you?" | cut -c 2-
```

This command will print the string, beginning with the second character, while removing the first character.

Remove First Four Characters from String

To remove the first four characters from the string “*hello, how are you?*” the command would be:

```
$ echo "hello, how are you?" | cut -c 5-
```

This command will print the string, beginning from the fifth character, while removing the first four characters.

Print String Between 2nd and 5th Characters

To print the string “*hello, how are you?*” between the second and fifth characters, the command would be:

```
$ echo "hello, how are you?" | cut -c 2-5
```

This command will print the string, beginning from the second character and up to the fifth character, while removing the remaining beginning and ending characters.

Remove Last Character from String

To remove the last character from the string *"hello, how are you?"* use the **cut** command with **rev**, as follows:

```
$ echo "hello, how are you?" | rev | cut -c2- | rev
```

This command works by first reversing the string, then cutting the first character, and finally reversing it again to give you the desired output.

Remove Last Four Characters from String

To remove the last four characters from the line *"hello, how are you?"* the command would be:

```
$ echo "hello, how are you?" | rev | cut -c5- | rev
```

This command works by first reversing the string, then cutting the first four characters, and then reversing it again to give you the desired output.

Remove First and Last Characters from String

To remove the first and last characters from the string *"hello, how are you?"* use the **cut** command with **rev**, as follows:

```
$ echo "hello world!" | cut -c2- | rev | cut -c2- | rev
```

This command works by cutting the first character, then reversing the string and cutting its first character, and then reversing it again to give you the desired output.

Remove Character from String Using tr

The **tr** command (short for translate) is used to translate, squeeze, and delete characters from a string. You can also use **tr** to remove characters from a string.

For demonstration purposes, we will use a sample string and then pipe it to the **tr** command.

Remove All Occurrences of Character

Using the **tr** command, you can remove all occurrences of lowercase or uppercase characters from your string. For instance, to remove all occurrences of the lowercase character 'h' from the string, the command would be:

```
$ echo "Hello, how are you?" | tr -d h
```

Similarly, to remove all occurrences of the uppercase character 'H' from the string, the command would be:

```
$ echo "Hello, how are you?" | tr -d H
```

You can also use interpreted sequences to remove lowercase or uppercase letters:

```
$ echo "Hello, how are you?" | tr -d [:upper:]
```

```
$ echo "HELLO, How are you?" | tr -d [:lower:]
```

Remove All Occurrences of Lowercase and Uppercase Characters

You can also remove all occurrences of both lowercase and uppercase characters from a string. For instance, the following command will remove all occurrences of the character 'h,' both lowercase and uppercase.

```
$ echo "Hello, how are you?" | tr -d 'hH'
```

Remove All Occurrences of Characters in a Specific Range

To remove all occurrences of characters from a string in the specific range 'd-h,' the command would be:

```
$ echo "hello, how are you?" | tr -d 'd-h'
```

This command will remove all characters in the range 'd-h' (d,e,f,g,h) in the string.

Conclusion

In Linux, there will always be more than one way to accomplish a simple job. The same is true with removing characters from a string. This article showed you four different ways to do so, along with a few examples for removing unwanted characters from a string. Deciding which tool to use all depends on your preferences and, more importantly, on what you want to achieve.

ABOUT THE AUTHOR



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